LAMP-Merge: Long-tail-Aware Medical Prototype Merging for One-shot Asynchronous Clinical Models

Anonymous submission

Abstract

Model merging aims to integrate multiple independently trained local checkpoints into a unified global model, typically through general-purpose paradigms such as parameter interpolation, task-vector composition, or update-direction alignment. These methods implicitly assume that client models share relatively complete and comparable global discriminative structures. We argue that this assumption does not hold in medical settings, giving rise to two diagnostic-structure anomalies: (1) Aggregation-induced prediction collapse after merging: When clients have incomplete diagnostic-class coverage, multiple locally effective models can be merged into a globally collapsed model whose predictive distribution concentrates on a small number of dominant classes. (2) Long-tailed medical prevalence cannot be ignored. Majority classes in medical data often reflect genuine clinical prevalence rather than merely data noise. To address these challenges, we propose LAMP-Merge, which recasts medical model merging from global parameter interpolation as class-level diagnostic-evidence reconstruction. Through diagnostic prototype reconstruction(DPR) and long-tail prevalence calibration(LPC), LAMP-Merge constructs a global diagnostic model without exposing raw medical images or requiring multi-round federated communication. Experiments on the Blood, Derma, Organ-C, Organ-S, and Ultrasound medical imaging datasets show that LAMP-Merge outperforms existing model-merging methods in most client-averaged settings while substantially mitigating prediction collapse.

Introduction

Federated learning (McMahan et al. 2017; Li et al. 2020; Karimireddy et al. 2020; Kairouz et al. 2021) and model merging (Izmailov et al. 2018; Wortsman et al. 2022; Ilharco et al. 2023; Yadav et al. 2023) provide two important collaboration routes for privacy-sensitive multi-center medical intelligence. Federated learning enables hospitals to train collaboratively without sharing raw data (Sheller et al. 2020; Rieke et al. 2020), while model merging further reduces communication and optimization costs by integrating independently trained local models after training (Ainsworth, Hayase, and Srinivasa 2023; Stoica et al. 2024; Wu et al. 2025). These paradigms are particularly attractive for medical imaging, because a single hospital rarely covers the full disease spectrum, while privacy regulations and device heterogeneity make centralizing raw medical images difficult (Kaissis et al. 2020a; Bonawitz et al. 2019; Abadi et al. 2016).

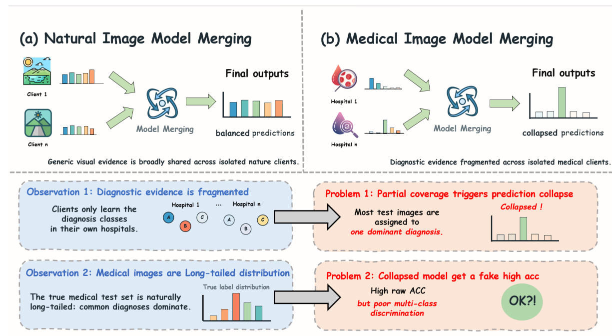


Figure 1: In multicenter healthcare settings, individual clients typically observe only a subset of the global diagnostic space. Conventional parameter-level model-merging methods can therefore cause the merged global model's predictions to concentrate on a small number of dominant classes.

Despite their success on natural images and general visual tasks (Izmailov et al. 2018; Wortsman et al. 2022; Ilharco et al. 2023; Entezari et al. 2022; Ainsworth, Hayase, and Srinivasa 2023), transferring these paradigms to medical data remains challenging. A key reason is that general-purpose model merging implicitly assumes that different client checkpoints contain relatively complete and comparable global discriminative structures; local models are expected to align through parameter space, task-vector space, update signs, or curvature statistics (Yadav et al. 2023; Yu et al. 2024; Wu et al. 2025; Jin et al. 2025). This assumption does not hold in medical scenarios. Medical centers differ not only in device domains but also in diagnostic-class availability: some centers may mainly cover common diseases or specific examination types, while having few rare-disease samples (Guan and Liu

2022; Zech et al. 2018; Li et al. 2021). The resulting local models are not complete replicas of the global task, but partial diagnostic experts supported by locally observed diseases.

As illustrated in Fig. 1, we summarize these challenges as two diagnostic-structure abnormalities in medical model merging:

- Diagnostic evidence fragmented across isolated medical clients. Each client has reliable supervision only for observed classes; thus, a local model may be locally discriminative but still lack evidence for the full global diagnostic space (Guan and Liu 2022; Li et al. 2021).
- Long-tailed Medical Image Prevalence. Medical image datasets naturally follow a long-tailed distribution: a few common diagnostic classes dominate the samples, while rare classes are underrepresented (Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010). Under this imbalanced class prior, a merged model collapsed to the majority class can still obtain high Accuracy despite lacking complete multi-class discrimination.

To address these issues, we propose LAMP-Merge, a long-tail-aware medical prototype-merging method for single-shot asynchronous medical image model merging. LAMP-Merge shifts the merge target from global parameter interpolation to class-level diagnostic-evidence reconstruction. The key idea is to use client-uploaded class-level statistics as a synchronization signal for the global diagnostic space, allowing the server to reconstruct a global diagnostic classifier without accessing raw medical images, using a server-side validation set, or conducting multi-round federated communication. Specifically, LAMP-Merge consists of two mechanisms: diagnostic prototype reconstruction aggregates reliable class-level evidence through prototypes in a shared reference feature space, and long-tail prevalence calibration injects a bounded prior bias from client-uploaded class counts, thereby mitigating prediction collapse while preserving clinically meaningful majority-class prevalence.

Our main contributions are summarized as follows:

- Medical Merging Diagnosis. We revisit model merging under single-shot asynchronous medical collaboration and identify limited local diagnostic coverage and aggregation-induced prediction collapse as two key abnormalities caused by medical class availability and long-tailed prevalence.
- LAMP-Merge. We propose a long-tail-aware medical prototype-merging method that reconstructs a global diagnostic classifier from client-side class prototypes and prevalence statistics without exposing raw images or requiring multi-round federated optimization.
- Empirical Validation. We conduct full-scope experiments on five medical image datasets, four vision-model families, three client-population sizes, and three non-IID skew levels, showing that LAMP-Merge consistently outperforms strong model-merging baselines and improves prediction-collapse diagnostics.

Related Works

Model Merging

Model-merging methods integrate independently trained checkpoints in parameter, task-vector, or geometric spaces. Weight averaging (Izmailov et al. 2018) and Model Soups (Wortsman et al. 2022) directly interpolate parameters; permutation-aware matching (Entezari et al. 2022; Ainsworth, Hayase, and Srinivasa 2023) aligns hidden-unit symmetries; ZipIt (Stoica et al. 2024) performs feature-level zipping; Task Arithmetic (Ilharco et al. 2023), TIES-Merging (Yadav et al. 2023), and DARE (Yu et al. 2024) manipulate task deltas and sign conflicts; Fisher merging (Wu et al. 2025), RegMean (Jin et al. 2025), Model Stock (Jang, Yun, and Han 2025), Breadcrumbs (Davari and Belilovsky 2024), Iso-C (Marczak et al. 2025), FreeMerge (Zheng and Wang 2025), and RobustMerge (Zeng et al. 2025) introduce curvature, mean-statistic, subspace, frequency-domain, or robustness assumptions.

However, these methods share a class-agnostic merging interface: their variables are global parameters, task vectors, update signs, or geometric statistics rather than class-level diagnostic evidence. They do not encode whether a client has reliable supervision for a specific medical class. Under incomplete local class coverage and long-tailed prevalence, reliable class directions can be mixed with absent or conflicting directions from other clients, resulting in aggregation-induced prediction collapse(Buda, Maki, and Mazurowski 2018; Zech et al. 2018).

One-shot clinical merging therefore requires class availability to be distinguished from model-wide parameter similarity.

Medical Model Merging

Medical fusion studies exploit domain structures such as irregular clinical time series, physiological sensor graphs, and image-EHR fusion. mTAN (Shukla and Marlin 2021), Raindrop (Zhang et al. 2022), and MedFuse (Hayat, Geras, and Shamout 2023) show that medical tasks benefit from modeling clinical sampling mechanisms, modality heterogeneity, and patient-level temporal relations.

However, their fusion objects are usually patient-level inputs, multimodal records, or clinical representations rather than independently trained image-classifier checkpoints. They often require patient samples, timestamps, electronic health records, or server-side validation feedback, which conflicts with common privacy constraints (Kaissis et al. 2020a; Sheller et al. 2020; Bonawitz et al. 2019). MedMerge (Wei et al.

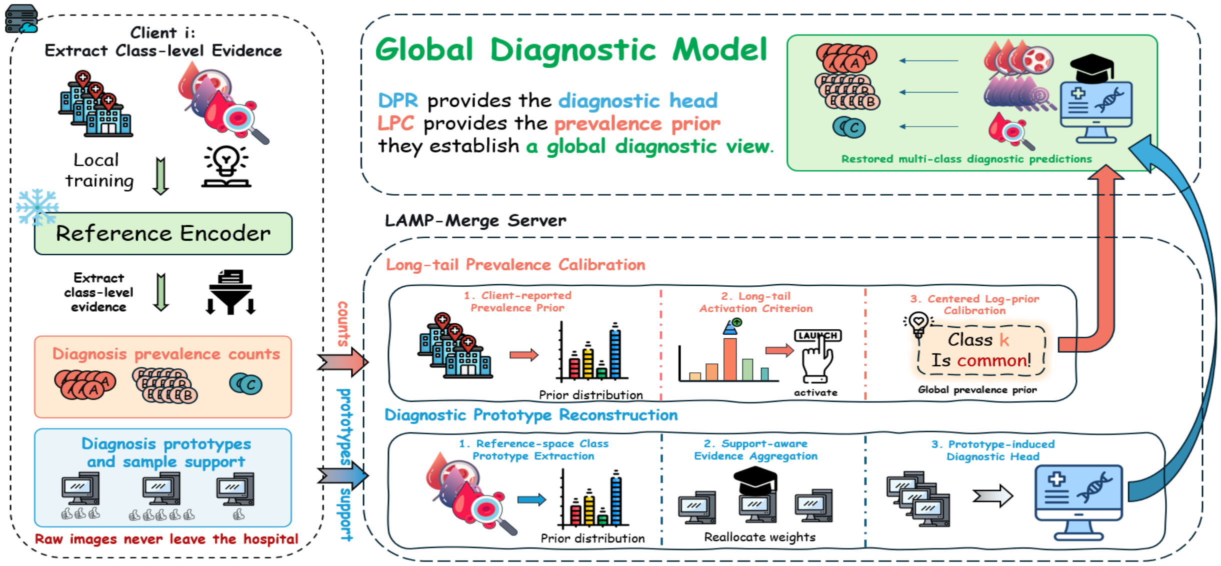


Figure 2: Local models from multicenter hospitals typically contain only partial diagnostic knowledge, while raw medical images cannot be centrally shared. LAMP-Merge reconstructs a privacy-friendly global medical classifier through DPR and LPC, without accessing raw data or requiring multi-round federated optimization.

2026) is a relevant training-free medical knowledge-transfer framework, but it targets cross-modal adaptation for medical vision-language models rather than asynchronous checkpoint merging for the same classification task. Thus, existing medical fusion routes are mismatched with our privacy and single-upload protocol constraints. LAMP-Merge keeps the medical structure uploadable by using only locally computed class prototypes and prevalence counts.

Federated Learning

Federated and decentralized learning perform collaborative training through communication protocols. FedAvg (McMahan et al. 2017) and its variants or medical deployments (Li et al. 2020; Karimireddy et al. 2020; Li et al. 2021; Kaissis et al. 2020b; Sheller et al. 2020; Rieke et al. 2020; Kairouz et al. 2021) rely on repeated server-client updates; Swarm Learning (Warnat-Herresthal et al. 2021) and Gossip Learning (Heged\Hus, Danner, and Jelasity 2019) reduce centralization but still require participants to continue training, exchanging models, and optimizing from intermediate states.

However, our setting begins after local training has finished. Hospitals may not remain online, receive global checkpoints repeatedly, or run additional optimization rounds. The bottleneck is therefore communication feasibility in addition to privacy. LAMP-Merge requires one upload of a local model interface and class-level statistics, enabling the server to construct a unified diagnostic model without a multi-round feedback loop.

Method

In this section, we first formulate the diagnostic-structure pathologies exposed when generic model merging is applied to medical data, namely limited local diagnostic coverage and aggregation-induced prediction collapse. To correct these intrinsic mismatches, we propose LAMP-Merge, a two-stage framework composed of DPR and LPC, as shown in Fig. 2.

Preliminaries

Problem Formulation. Assume K medical clients. Client i owns a private labeled dataset $D_i = \{(x, y)\}$ drawn from a class-skewed medical distribution over C diagnosis classes. Let $D_{i,c}$ denote the subset of client i belonging to diagnosis class c. In medical partitions, $D_{i,c}$ can be empty for many classes; therefore, client i has reliable supervision only for its observed classes. The goal is to construct a global diagnostic model under this partial and heterogeneous diagnostic-label coverage.

Upload and Server Constraints. After local training, each client uploads its checkpoint interface and class-level statistics once; the server receives no raw images, per-sample data, validation set, or iterative feedback. The uploaded support $n_{i,c}$, prevalence $m_{i,c}$, and reference-space prototype $\mu_{i,c}$ enable one merge with class-level diagnostic evidence as an explicit variable.

Motivation

Although generic model merging is training-free and parameter-efficient, we find that its diagnostic structure can severely degenerate in medical model merging. Prediction-distribution diagnosis and class-support analysis reveal two key abnormalities:

(1) Aggregation-induced prediction collapse. Under partial diagnostic coverage, parameter-level merging can turn several locally useful predictors into a globally collapsed predictor. The merged model tends to allocate most test images to a few dominant classes, indicating that global class-discriminative structure has not been reconstructed.

(2) Prevalence prior cannot be ignored. Medical datasets often exhibit clinically meaningful long-tailed prevalence. Therefore, completely suppressing majority-class dominance is also inappropriate: a collapsed predictor is undesirable, but a merger should still preserve calibrated disease prevalence when the dominant class reflects the true clinical prior.

Diagnostic Prototype Reconstruction Module

To address fragmented diagnostic evidence across isolated medical clients, DPR reconstructs class-conditional evidence in a shared reference space and synthesizes a global diagnostic head over the full label space.

(1) Reference-space Class Prototype Extraction. First, all clients extract class prototypes with the same frozen reference backbone. Let ϕ_0 be the reference backbone instantiated from the shared model configuration before local training, and let $T(\cdot)$ be the evaluation transform. For each diagnosis class c, client i locally computes a reference-space class prototype

$$\mu_{i,c} = \frac{1}{n_{i,c}} \sum_{(x,y)\in D_i} \mathbf{1}[y=c]\phi_0(T(x)), \tag{1}$$

where $n_{i,c} = |D_{i,c}|$ is the class support count. If $n_{i,c} = 0$, the client uploads zero support for that class and contributes no prototype evidence.

(2) Support-aware Evidence Aggregation. Second, the server converts class-support counts into evidence weights to characterize the reliability of each client for a given class:

$$e_{i,c} = (n_{i,c}+1)^{\gamma}\mathbf{1}[n_{i,c} > 0], \tag{2}$$

where $\gamma = 0.55$ is the default evidence exponent. This weight gives larger class-specific contribution to clients with more supervised samples of class c, while excluding clients that never observe this class. The normalized reliability is

$$\alpha_{i,c} = \frac{e_{i,c}}{\sum_{j=1}^{K} e_{j,c}}. \tag{3}$$

The global diagnosis prototype is obtained by class-wise reliability-weighted aggregation:

$$p_c = \sum_{i=1}^{K} \alpha_{i,c}\mu_{i,c}. \tag{4}$$

This step aggregates evidence only from clients that have actually observed class c, preventing absent-class classifiers from diluting valid diagnostic signals.

(3) Prototype-induced Diagnostic Head. Finally, LAMP-Merge maps global diagnosis prototypes into a cosine prototype classifier:

$$w_c = s\frac{p_c}{\|p_c\|_2}, \tag{5}$$

where $s = 18.75$ is the default head scale. The resulting head is not an average of local classifier parameters; it is induced by class evidence in the shared reference space, thereby restoring multiclass diagnostic directions over the global label space.

Long-tail Prevalence Calibration

To address long-tailed medical prevalence cannot be ignored, LPC adds a bounded class prior after DPR to calibrate the clinical prevalence reflected by client statistics.

(1) Client-reported Prevalence Prior. First, each client uploads class-prevalence counts $m_{i,c}$. The server estimates the global class prior

$$\pi_c = \frac{\sum_{i=1}^{K} m_{i,c}}{\sum_{k=1}^{C}\sum_{i=1}^{K} m_{i,k}}. \tag{6}$$

This prior is computed only from class-level counts, not from raw images, per-sample features, or per-sample predictions.

(2) Long-tail Activation Criterion. Second, to avoid unnecessary calibration in non-long-tailed cases, the server measures dominant-class imbalance by

$$r = C\max_c \pi_c. \tag{7}$$

The calibration module is activated only when $r > \tau$, where $\tau = 2.5$ by default.

(3) Centered Log-prior Calibration. Finally, when the module is activated, LAMP-Merge constructs a centered log-prior bias

$$b_c = \lambda\left(\log\pi_c - \frac{1}{C}\sum_{k=1}^{C}\log\pi_k\right), \tag{8}$$

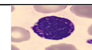
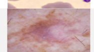
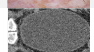
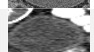
with default maximum calibration strength $\lambda = 4.25$. The centering term prevents a uniform shift of all class logits, so the bias only changes the relative class prior. This module calibrates prevalence without replacing the diagnostic directions reconstructed by the first module.

Optimization Objective and Algorithm

The final scoring function of LAMP-Merge integrates the diagnostic prototype reconstruction term and the long-tail prevalence calibration term:

$$\ell_c(x) = w_c^{\top}\phi_0(T(x)) + \mathbf{1}[r > \tau]\,b_c. \tag{9}$$

Table 1: Dataset statistics and representative samples for the five medical image benchmarks. Counts are taken from the exact NPZ files used in our experiments.

Dataset	Classes	Train	Val	Test	Total	Sample
Blood	8	11,959	1,712	3,421	17,092	
Derma	7	7,007	1,003	2,005	10,015	
Organ-C	11	12,975	2,392	8,216	23,583	
Organ-S	11	13,932	2,452	8,827	25,211	
Ultrasound	8	3,869	549	1,113	5,531	

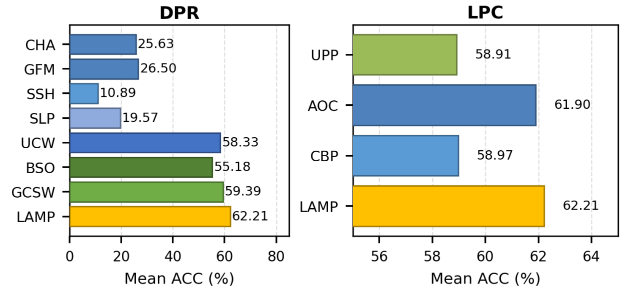


Figure 3: Internal ablations of DPR and LPC. Definitions of the plotted abbreviations are provided in the supplementary ablation configurations.

Experiments

Experimental Settings

(1) Evaluation Datasets: We evaluate the merged models on five representative MedMNIST v2 benchmarks(Yang, Shi, and Ni 2023): Blood, Derma, Organ-C, Organ-S, and Ultrasound. They cover blood-cell microscopy, dermoscopy, abdominal CT, and ultrasound image classification (detailed dataset descriptions are provided in the Appendix).

(2) Baselines: We compare LAMP-Merge with twelve established merging baselines spanning parameter averaging, task-vector pruning, conflict resolution, statistic-aware correction, interpolation, subspace merging, Fourier fusion, and robust merging (Yadav et al. 2023; Yu et al. 2024; Jin et al. 2025; Wu et al. 2025; Davari and Belilovsky 2024; Jang, Yun, and Han 2025; Ilharco et al. 2023; Marczak et al. 2025; Zheng and Wang 2025; Zeng et al. 2025).

(3) Merging Settings: We evaluate four vision-model families: ResNet, ConvNeXt, ViT-Tiny, and Swin-Tiny(He et al. 2016; Liu et al. 2022; Dosovitskiy et al. 2021; Liu et al. 2021). For each dataset-model pair, we construct non-IID client partitions with $K \in \{3, 5, 7\}$ clients and Dirichlet skew parameter $\beta \in \{0, 0.01, 0.1\}$, using seed 42(Hsu, Qi, and Brown 2019). A client-average result fixes the dataset, model, and K, and then averages over the three β values. And the experiments follow a single-shot asynchronous model-merging protocol.

Comparison With the State of the Arts

Tables 2–5 report the full client-average results for all four backbones. Across all evaluated settings, LAMP-Merge wins or ties the strongest non-LAMP baseline in 57 cells (95.0%).

Ablation Study

To assess DPR and LPC, we conduct module ablations under the main experimental configuration. Weight averaging, weight averaging + DPR, and full LAMP-Merge obtain mean ACC of 22.04%, 58.91%, and 62.21%, respectively; Fig. 4 further reports strict 2×2 TIES/DARE controls.

1) Effectiveness of DPR To identify the necessary components of DPR, we retain LPC and replace its prototype semantics with classifier-head aggregation (CHA), global-feature mean (GFM), support-only synthetic head (SSH), or shuffled-label prototype (SLP). These settings test head directions, class conditioning, synthetic directions, and label–prototype alignment. We also replace its client-support weighting with uniform client weight (UCW), binary support only (BSO), or global client-size weight (GCSW), which test the reliability assigned to each observed class.

Table 6 and Fig. 3 show that every replacement lowers ACC. This confirms that DPR needs both reliable class prototypes and class-specific support weighting.

2) Effectiveness of LPC To evaluate LPC, we retain DPR and compare its empirical prior with the uniform prevalence prior (UPP), always-on calibration (AOC), and client-balanced prior (CBP) across the full experimental setting. UPP tests whether measured medical prevalence is useful, AOC tests the activation gate, and CBP tests whether equal client influence can replace sample-weighted prevalence.

Table 7 and Fig. 3 show that LAMP-Merge is best and every control reduces ACC. This demonstrates that calibrated empirical prevalence and client-size-aware aggregation complement DPR.

Prediction-collapse Diagnostics

Because raw ACC can hide single-class predictions on long-tailed data, we additionally report collapse ratio, effective predicted classes, and predicted–true total variation(Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010).

Table 8 shows that LAMP-Merge consistently improves all collapse diagnostics.

Visualization Experiment

We visualize the absolute per-class predicted–true gap; values closer to zero indicate better prevalence align-

Table 2: Client-average test accuracy (%) on ResNet. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.	33.45	24.67	28.22	28.78	62.91	47.00	62.44	57.45	22.61	17.72	13.96	18.10	28.64	17.94	24.75	23.78	24.35	15.81	15.42	18.53
TIES[NeurIPS 2023]	26.01	18.72	17.87	20.87	59.95	48.41	64.84	57.73	26.50	16.84	21.18	21.51	19.81	19.15	18.74	19.23	34.77	19.89	18.54	24.40
DARE-Linear[ICML 2024]	28.32	28.56	23.92	26.93	63.57	48.41	62.88	58.29	20.18	25.88	16.50	20.85	25.45	22.45	25.37	24.42	21.62	17.28	16.08	18.33
DARE-TIES[ICML 2024]	31.94	19.35	17.80	23.03	53.75	48.25	55.58	52.53	18.51	15.15	18.59	17.42	17.90	13.52	17.30	16.24	29.32	16.35	18.90	21.52
RegMean[ICLR 2023]	34.60	23.69	34.07	30.79	58.12	28.43	50.22	45.59	24.05	13.40	13.88	17.11	25.04	16.23	14.93	18.73	22.76	16.95	13.18	17.63
Fisher[NeurIPS 2022]	31.96	22.37	26.08	26.80	67.07	67.02	66.90	67.00	29.73	17.41	18.12	21.75	20.10	29.56	19.63	23.10	23.33	18.60	25.13	22.35
Breadcrumbs[ECCV 2024]	8.54	14.17	14.60	12.44	4.95	29.13	17.21	17.10	7.66	6.76	6.85	7.09	8.02	9.48	6.03	7.84	12.25	12.73	10.21	11.73
Model Stock[ECCV 2024]	24.20	18.24	19.75	20.73	62.16	43.11	57.21	54.16	12.25	12.74	11.85	12.28	27.42	15.78	20.12	21.11	17.91	13.21	15.78	15.63
Iso-C[ICML 2025]	32.78	27.48	30.23	30.16	50.49	44.22	64.16	52.96	23.31	16.29	13.83	17.81	27.36	21.18	24.53	24.36	26.15	16.98	15.54	19.56
Free-Merging[ICCV 2025]	33.45	24.67	28.22	28.78	62.91	47.00	62.46	57.46	22.61	17.71	13.96	18.09	28.63	17.93	24.75	23.77	24.35	15.81	15.45	18.54
RobustMerge[NeurIPS 2025]	34.01	24.30	22.56	26.96	29.56	20.85	39.42	29.94	24.58	17.64	16.53	19.58	26.80	22.27	23.01	24.03	21.89	22.31	17.07	20.42
FROM[AAAI 2026]	29.89	23.06	20.52	24.49	67.13	43.69	65.80	58.87	23.50	15.60	15.20	18.10	22.07	20.87	24.76	22.57	23.09	15.06	20.63	19.59
LAMP-Merge	80.01	79.94	79.92	79.95	67.55	67.63	67.63	67.60	60.05	59.94	60.07	60.02	52.98	52.79	52.81	52.86	48.07	46.75	47.35	47.39

Table 3: Client-average test accuracy (%) on ConvNeXt. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.	15.65	11.22	13.84	13.57	29.66	48.30	66.88	48.28	10.37	12.18	9.44	10.66	13.08	10.35	14.16	12.53	15.06	15.06	12.52	14.21
TIES[NeurIPS 2023]	17.15	12.14	15.24	14.84	29.66	17.97	27.13	24.92	12.17	7.86	7.16	9.06	11.23	7.61	7.96	8.93	15.18	14.56	12.91	14.22
DARE-Linear[ICML 2024]	17.56	11.22	13.45	14.08	27.05	48.30	48.30	41.22	10.35	6.99	10.85	9.40	12.82	10.35	7.97	10.38	15.75	15.75	12.64	14.71
DARE-TIES[ICML 2024]	10.96	15.19	14.97	13.71	9.08	27.98	24.51	20.52	13.05	13.46	7.70	11.40	10.96	9.46	7.09	9.17	14.56	12.97	15.00	14.18
RegMean[ICLR 2023]	15.65	8.17	18.56	14.13	26.33	7.73	27.20	20.42	11.43	10.17	7.68	9.76	10.14	12.59	10.38	11.04	15.75	15.39	15.00	15.38
Fisher[NeurIPS 2022]	15.65	9.99	11.89	12.51	48.25	26.38	29.71	34.78	10.03	9.74	9.15	9.64	6.04	5.17	8.43	6.55	16.95	13.60	14.56	15.04
Breadcrumbs[ECCV 2024]	15.16	13.66	11.63	13.48	5.24	27.66	46.30	26.40	10.31	14.44	15.74	13.50	15.16	15.54	20.77	17.16	13.84	13.63	13.63	13.70
Model Stock[ECCV 2024]	17.56	11.44	17.78	15.59	29.66	48.30	66.88	48.28	12.76	12.18	13.85	12.93	13.08	10.35	19.35	14.26	15.75	12.91	14.79	14.48
Iso-C[ICML 2025]	19.47	18.65	14.80	17.64	29.71	48.30	66.88	48.30	9.91	6.99	9.45	8.78	7.89	14.77	8.67	10.44	15.18	12.91	16.05	14.71
Free-Merging[ICCV 2025]	15.75	11.90	13.84	13.83	29.71	48.30	66.88	48.30	9.06	12.18	13.85	11.70	13.08	10.35	14.16	12.53	15.06	12.91	16.95	14.97
RobustMerge[NeurIPS 2025]	14.10	14.05	7.52	11.89	9.08	45.07	48.30	34.15	13.94	12.18	17.93	14.68	23.54	9.61	20.77	17.97	13.00	11.35	13.63	12.66
FROM[AAAI 2026]	15.65	10.83	16.71	14.40	29.66	34.06	45.69	36.47	12.32	12.24	6.50	10.35	13.24	11.05	8.97	11.09	15.75	15.78	10.87	14.13
LAMP-Merge	84.85	84.87	84.98	84.90	59.68	58.77	59.42	59.29	65.41	65.33	65.51	65.42	60.55	60.50	60.66	60.57	43.07	42.80	42.41	42.76

Table 4: Client-average test accuracy (%) on ViT-Tiny. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.	13.43	10.82	11.60	11.95	46.38	43.69	36.96	42.34	16.57	10.43	9.52	12.17	8.44	8.52	17.10	11.35	14.50	13.27	18.84	15.54
TIES[NeurIPS 2023]	7.96	14.21	12.85	11.67	47.80	34.98	46.15	42.98	13.41	12.23	9.53	11.72	11.92	10.02	15.97	12.64	20.01	13.06	10.18	14.42
DARE-Linear[ICML 2024]	23.97	11.29	13.87	16.38	28.50	10.02	12.04	16.85	15.34	9.46	7.04	10.61	7.48	5.49	16.11	9.69	13.66	10.51	14.35	12.84
DARE-TIES[ICML 2024]	7.35	11.89	11.46	10.23	27.35	45.02	35.28	35.88	12.52	9.03	10.33	10.63	5.41	8.74	5.91	6.69	14.47	11.20	10.42	12.03
RegMean[ICLR 2023]	17.48	14.17	12.90	14.85	51.14	36.67	66.58	51.46	12.50	9.64	9.83	10.66	12.63	7.17	10.68	10.16	14.91	13.84	11.44	13.40
Fisher[NeurIPS 2022]	11.66	15.12	19.44	15.41	47.80	20.42	48.20	38.81	7.99	10.72	14.24	10.98	8.56	12.18	12.22	10.99	16.11	13.15	14.20	14.49
Breadcrumbs[ECCV 2024]	13.63	10.39	13.00	12.34	4.89	25.72	6.07	12.23	7.73	10.00	16.44	11.39	12.37	4.83	12.19	9.80	13.60	11.41	15.24	13.42
Model Stock[ECCV 2024]	19.40	11.20	11.34	13.98	38.05	30.79	27.35	32.06	14.88	12.27	12.15	13.10	8.93	8.59	18.67	12.06	15.99	18.09	16.41	16.83
Iso-C[ICML 2025]	12.34	14.83	10.18	12.45	57.09	45.29	23.06	41.81	10.84	5.67	8.97	8.49	6.11	5.31	10.81	7.41	14.17	15.27	18.69	16.04
Free-Merging[ICCV 2025]	19.89	14.57	13.92	16.13	46.30	47.70	38.55	44.18	14.25	9.80	7.59	10.55	7.27	9.14	16.43	10.95	14.05	15.93	16.47	15.48
RobustMerge[NeurIPS 2025]	14.82	10.70	14.25	13.26	3.47	27.68	17.24	16.13	9.54	6.17	14.38	10.03	12.60	7.07	12.17	10.61	16.98	18.54	15.63	17.05
FROM[AAAI 2026]	18.02	7.81	14.05	13.29	46.95	51.21	66.88	55.01	21.48	13.01	9.72	14.74	7.72	7.98	14.85	10.18	18.36	17.43	17.76	17.85
LAMP-Merge	85.21	85.06	84.98	85.09	59.45	59.32	59.83	59.53	58.78	58.40	58.29	58.49	54.39	54.51	54.65	54.52	46.12	45.97	45.94	46.01

Table 5: Client-average test accuracy (%) on Swin-Tiny. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.	15.65	11.86	15.24	14.25	45.69	48.30	66.88	53.62	10.68	12.94	7.87	10.50	6.82	6.04	13.81	8.89	15.27	12.94	14.65	14.29
TIES[NeurIPS 2023]	9.72	11.90	13.84	11.82	8.41	7.80	66.88	27.70	11.60	7.02	9.34	9.32	7.34	6.07	11.07	8.16	16.95	13.03	12.61	14.20
DARE-Linear[ICML 2024]	15.65	11.61	18.25	15.17	45.69	27.71	46.30	39.90	11.55	12.72	7.76	10.68	6.40	7.20	13.81	9.14	17.43	15.36	14.97	15.92
DARE-TIES[ICML 2024]	13.84	11.62	13.84	13.10	29.61	26.38	66.88	40.96	12.76	7.74	11.16	10.55	11.94	4.34	12.30	9.53	17.91	13.51	13.00	14.81
RegMean[ICLR 2023]	13.74	17.15	15.78	15.56	27.05	46.30	45.69	39.68	14.11	7.73	8.11	9.98	12.15	8.21	9.26	9.87	14.65	11.05	17.79	14.50
Fisher[NeurIPS 2022]	16.33	17.56	15.65	16.51	45.82	6.52	47.18	33.17	8.14	8.99	9.78	8.97	6.51	6.50	10.92	7.98	17.46	11.65	17.55	15.55
Breadcrumbs[ECCV 2024]	15.00	13.65	15.37	14.67	6.47	27.71	7.15	13.78	14.45	17.83	16.60	16.29	23.54	20.77	21.11	21.81	14.50	13.12	16.53	14.72
Model Stock[ECCV 2024]	13.74	14.40	17.15	15.10	45.69	27.71	48.25	40.55	11.10	7.71	14.04	10.95	6.71	11.06	17.14	11.64	15.39	12.73	18.69	15.60
Iso-C[ICML 2025]	14.30	17.22	17.00	16.17	30.99	28.40	51.74	37.04	13.02	13.27	14.69	13.66	15.45	17.27	18.80	17.17	17.55	17.31	17.91	17.59
Free-Merging[ICCV 2025]	11.54	8.61	18.40	12.85	45.69	48.30	66.88	53.62	9.59	13.77	8.90	10.75	6.82	4.97	10.62	7.47	15.18	15.75	15.99	15.64
RobustMerge[NeurIPS 2025]	13.74	14.09	16.70	14.84	23.77	27.71	5.04	18.84	16.01	14.42	19.60	16.68	16.73	11.51	20.77	16.34	17.61	13.15	16.29	15.68
FROM[AAAI 2026]	15.65	14.95	16.71	15.77	45.69	29.71	66.88	47.43	11.74	12.20	10.51	11.48	11.33	10.16	13.80	11.76	15.39	12.52	14.76	14.22
LAMP-Merge	77.38	77.34	77.17	77.30	63.24	63.18	63.09	63.17	68.49	68.17	68.12	68.26	62.58	62.80	62.84	62.74	47.71	48.70	48.40	48.27

Table 6: DPR internal ablation by dataset. Each entry averages all backbones, client counts, and Dirichlet settings; Avg averages the five datasets.

Category	Setting	Blood	Derma	Organ-C	Organ-S	US	Avg
Prototype semantics	CHA	18.95	56.14	17.74	18.88	16.43	25.63
	GFM	8.88	66.88	22.33	23.54	10.87	26.50
	SSH	11.62	17.29	7.08	7.74	10.69	10.89
	SLP	24.46	28.73	14.18	16.25	14.25	19.57
Client-support weighting	UCW	78.03	59.61	58.36	53.86	41.79	58.33
	BSO	78.03	44.96	57.86	53.24	41.79	55.18
	GCSW	78.45	62.48	59.11	53.79	43.13	59.39
	LAMP	81.81	62.40	63.05	57.67	46.11	62.21

Table 7: LPC internal ablation by dataset. UPP uses a uniform prior, AOC removes the activation gate, and CBP averages client-normalized priors. Each entry averages all backbones, client counts, and Dirichlet settings; Avg averages the five datasets.

Setting	Blood	Derma	Organ-C	Organ-S	US	Avg
UPP	81.81	46.58	62.57	57.50	46.11	58.91
AOC	80.57	62.40	63.05	57.67	45.80	61.90
CBP	81.25	49.61	62.10	57.12	44.76	58.97
LAMP-Merge	81.81	62.40	63.05	57.67	46.11	62.21

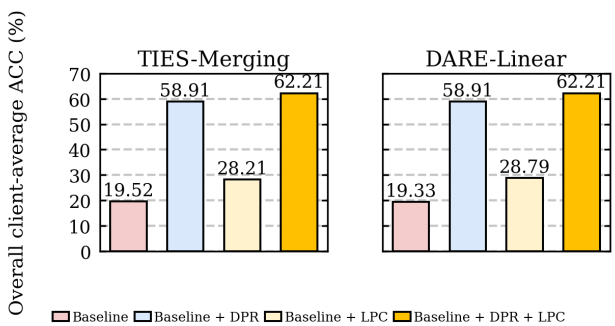


Figure 4: Strict 2×2 DPR/LPC ablations for TIES-Merging and DARE-Linear. All experimental datasets are included.

ment.

Hyperparameter Analysis Experiment

Fig. 6 evaluates DPR/LPC sensitivity on the full setting.

This single-factor ablation varies DPR's prototype-head scale s under several γ values and LPC's calibration strength λ under several τ values, while holding all remaining settings fixed. The selected $s = 18.75$ lies in a broad high-performing DPR region, and the closely spaced LPC curves around $\tau = 2.5$ show gradual sensitivity rather than an isolated peak.

Table 8: Prediction-collapse diagnostics by dataset. Best ref. is selected separately for each metric and dataset. Ratio metrics use percentage scale; Avg averages the five datasets.

Metric	Series	Blood	Derma	Organ-C	Organ-S	US	Avg
Collapse ↓	Best ref.	80.20	85.18	72.38	71.12	80.31	79.02
	LAMP	19.58	68.05	20.72	24.36	20.45	30.63
Eff. classes ↑	Best ref.	1.81	1.65	2.49	2.53	1.98	2.05
	LAMP	7.48	3.10	9.55	8.44	7.44	7.20
Pred-true TV ↓	Best ref.	73.48	46.03	75.42	73.85	72.73	69.44
	LAMP	3.97	17.09	12.38	13.17	15.51	12.42

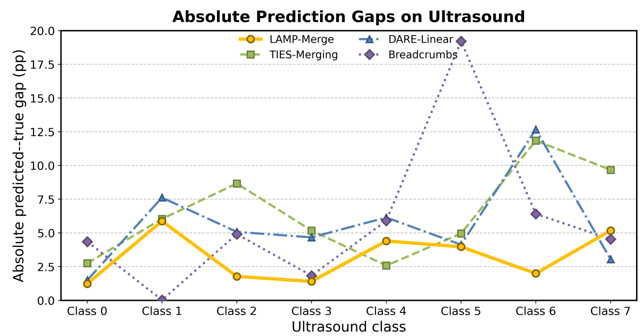


Figure 5: Absolute predicted–true class gaps on Ultrasound, averaged over all backbones, client counts, and Dirichlet settings. Values are percentage points; zero denotes exact agreement.

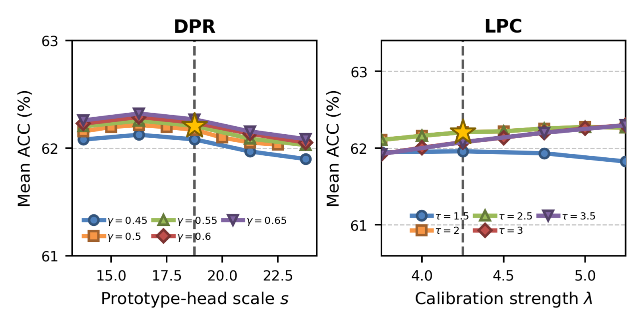


Figure 6: Single-factor DPR/LPC ablation on the full setting; each curve varies one module parameter while all others remain fixed.

Conclusion

We study one-shot medical model merging without raw-image sharing or federated rounds. Under incomplete local coverage, generic merges can collapse predictions, while raw ACC may conceal failure on long-tailed data. Across full-scope experiments, DPR restores class-level directions and LPC retains clinically meaningful prevalence without reintroducing collapse. Together, they offer a class-level interface for privacy-constrained collaboration.

References

Abadi, M.; Chu, A.; Goodfellow, I.; McMahan, H. B.; Mironov, I.; Talwar, K.; and Zhang, L. 2016. Deep Learning with Differential Privacy. In Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security, 308–318.

Ainsworth, S. K.; Hayase, J.; and Srinivasa, S. 2023. Git Re-Basin: Merging Models modulo Permutation Symmetries. arXiv:2209.04836.

Bonawitz, K.; et al. 2019. Towards Federated Learning at Scale: System Design. In Proceedings of Machine Learning and Systems, volume 1, 374–388.

Brodersen, K. H.; Ong, C. S.; Stephan, K. E.; and Buhmann, J. M. 2010. The Balanced Accuracy and Its Posterior Distribution. In 2010 20th International Conference on Pattern Recognition, 3121–3124.

Buda, M.; Maki, A.; and Mazurowski, M. A. 2018. A systematic study of the class imbalance problem in convolutional neural networks. Neural Networks, 106: 249–259.

Davari, M.; and Belilovsky, E. 2024. Model Breadcrumbs: Scaling Multi-Task Model Merging with Sparse Masks. arXiv:2312.06795.

Dosovitskiy, A.; et al. 2021. An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale. In International Conference on Learning Representations.

Entezari, R.; Sedghi, H.; Saukh, O.; and Neyshabur, B. 2022. The Role of Permutation Invariance in Linear Mode Connectivity of Neural Networks. In International Conference on Learning Representations.

Guan, H.; and Liu, M. 2022. Domain Adaptation for Medical Image Analysis: A Survey. IEEE Transactions on Biomedical Engineering, 69(3): 1173–1185.

Hayat, N.; Geras, K. J.; and Shamout, F. E. 2023. MedFuse: Multi-modal fusion with clinical time-series data and chest X-ray images. arXiv:2207.07027.

He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep Residual Learning for Image Recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 770–778.

Heged\Hus, I.; Danner, G.; and Jelasity, M. 2019. Gossip Learning as a Decentralized Alternative to Federated Learning. In Pereira, J.; and Ricci, L., eds., Distributed Applications and Interoperable Systems, 74–90. Cham: Springer International Publishing. ISBN 978-3-030-22496-7.

Hsu, T.-M. H.; Qi, H.; and Brown, M. 2019. Measuring the Effects of Non-Identical Data Distribution for Federated Visual Classification. arXiv:1909.06335.

Ilharco, G.; Ribeiro, M. T.; Wortsman, M.; Gururangan, S.; Schmidt, L.; Hajishirzi, H.; and Farhadi, A. 2023. Editing Models with Task Arithmetic. arXiv:2212.04089.

Izmailov, P.; Podoprikhin, D.; Garipov, T.; Vetrov, D. P.; and Wilson, A. G. 2018. Averaging Weights Leads to Wider Optima and Better Generalization. In Proceedings of the 34th Conference on Uncertainty in Artificial Intelligence, 876–885.

Jang, D.-H.; Yun, S.; and Han, D. 2025. Model Stock: All we need is just a few fine-tuned models. arXiv:2403.19522.

Jin, X.; Ren, X.; Preotiuc-Pietro, D.; and Cheng, P. 2025. Dataless Knowledge Fusion by Merging Weights of Language Models. arXiv:2212.09849.

Kairouz, P.; McMahan, H. B.; Avent, B.; et al. 2021. Advances and Open Problems in Federated Learning. Foundations and Trends in Machine Learning, 14(1–2): 1–210.

Kaissis, G. A.; Makowski, M. R.; Rückert, D.; and Braren, R. F. 2020a. Secure, privacy-preserving and federated machine learning in medical imaging. Nature Machine Intelligence, 2(6): 305–311.

Kaissis, G. A.; Makowski, M. R.; Rückert, D.; and Braren, R. F. 2020b. Secure, privacy-preserving and federated machine learning in medical imaging. Nature Machine Intelligence, 2(6): 305–311.

Karimireddy, S. P.; Kale, S.; Mohri, M.; Reddi, S.; Stich, S.; and Suresh, A. T. 2020. SCAFFOLD: Stochastic Controlled Averaging for Federated Learning. In Proceedings of the 37th International Conference on Machine Learning, 5132–5143.

Li, T.; Sahu, A. K.; Talwalkar, A.; and Smith, V. 2020. Federated Optimization in Heterogeneous Networks. In Proceedings of Machine Learning and Systems, volume 2, 429–450.

Li, X.; Jiang, M.; Zhang, X.; Kamp, M.; and Dou, Q. 2021. Federated Learning on Non-IID Features via Local Batch Normalization. In International Conference on Learning Representations.

Liu, Z.; Mao, H.; Wu, C.-Y.; Feichtenhofer, C.; Darrell, T.; and Xie, S. 2022. A ConvNet for the 2020s. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 11976–11986.

Liu, Z.; et al. 2021. Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows. In Proceedings of the IEEE/CVF International Conference on Computer Vision, 10012–10022.

Marczak, D.; Magistri, S.; Cygert, S.; Twardowski, B.; Bagdanov, A. D.; and van de Weijer, J. 2025. No Task Left Behind: Isotropic Model Merging with Common and Task-Specific Subspaces. arXiv:2502.04959.

McMahan, B.; Moore, E.; Ramage, D.; Hampson, S.; and Arcas, B. A. y. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In Singh, A.; and Zhu, J., eds., Proceedings of the 20th International Conference on Artificial Intelligence and Statistics, volume 54 of Proceedings of Machine Learning Research, 1273–1282. PMLR.

Rieke, N.; Hancox, J.; Li, W.; Milletarì, F.; Roth, H. R.; Albarqouni, S.; Bakas, S.; Galtier, M. N.; Landman, B. A.; Maier-Hein, K.; Ourselin, S.; Sheller, M.; Summers, R. M.; Trask, A.; Xu, D.; Baust, M.; and Cardoso,

M. J. 2020. The future of digital health with federated learning. npj Digital Medicine, 3(1): 171.

Sheller, M. J.; Edwards, B.; Reina, G. A.; Martin, J.; Pati, S.; Kotrotsou, A.; Milchenko, M.; Xu, W.; Marcus, D.; Colen, R. R.; and Bakas, S. 2020. Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data. Scientific Reports, 10(1): 10933.

Shukla, S. N.; and Marlin, B. M. 2021. Multi-Time Attention Networks for Irregularly Sampled Time Series. arXiv:2101.10318.

Stoica, G.; Bolya, D.; Bjorner, J.; Ramesh, P.; Hearn, T.; and Hoffman, J. 2024. ZipIt! Merging Models from Different Tasks without Training. In International Conference on Learning Representations.

Warnat-Herresthal, S.; Schultze, H.; Shastry; and so on. 2021. Swarm Learning for decentralized and confidential clinical machine learning. Nature, 594(7862): 265–270.

Wei, Y.; Zhu, Q.; Xia, Z.; Wang, L.; Zou, Y.; Li, Y.; Luo, Y.; and Du, B. 2026. MedMerge: A Training-Free Model Merging Framework for Medical Knowledge Transfer into Vision–Language Models. IEEE Transactions on Radiation and Plasma Medical Sciences, 1–1.

Wortsman, M.; Ilharco, G.; Gadre, S. Y.; Roelofs, R.; Gontijo-Lopes, R.; Morcos, A. S.; Namkoong, H.; Farhadi, A.; Carmon, Y.; Kornblith, S.; and Schmidt, L. 2022. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In Chaudhuri, K.; Jegelka, S.; Song, L.; Szepesvari, C.; Niu, G.; and Sabato, S., eds., Proceedings of the 39th International Conference on Machine Learning, volume 162 of Proceedings of Machine Learning Research, 23965–23998. PMLR.

Wu, J.; Yang, J.; Wang, H.; Wen, J.; and Yu, Z. 2025. From Parameter to Representation: A Closed-Form Approach for Controllable Model Merging. ArXiv:2511.10943 [cs].

Yadav, P.; Tam, D.; Choshen, L.; Raffel, C.; and Bansal, M. 2023. TIES-Merging: Resolving Interference When Merging Models. arXiv:2306.01708.

Yang, J.; Shi, R.; and Ni, B. 2023. MedMNIST v2: A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification. Scientific Data, 10(1): 41.

Yu, L.; Yu, B.; Yu, H.; Huang, F.; and Li, Y. 2024. Language Models are Super Mario: Absorbing Abilities from Homologous Models as a Free Lunch. arXiv:2311.03099.

Zech, J. R.; Badgeley, M. A.; Liu, M.; Costa, A. B.; Titano, J. J.; and Oermann, E. K. 2018. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. PLOS Medicine, 15(11): 1–17.

Zeng, F.; Guo, H.; Zhu, F.; Shen, L.; and Tang, H. 2025. RobustMerge: Parameter-Efficient Model Merging for MLLMs with Direction Robustness. arXiv:2502.17159.

Zhang, X.; Zeman, M.; Tsiligkaridis, T.; and Zitnik, M. 2022. Graph-Guided Network for Irregularly Sampled Multivariate Time Series. arXiv:2110.05357.

Zheng, S.; and Wang, H. 2025. FREE-Merging: Fourier Transform for Efficient Model Merging. arXiv:2411.16815.

Supplementary Results and Analysis

Detailed Dataset Description

The experiments use five MedMNIST v2 benchmarks (Yang, Shi, and Ni 2023). They were selected to cover microscopy, dermoscopy, CT, and ultrasound while varying the number of diagnostic classes and the prevalence profile. Table 1 reports the exact split sizes used by the experiments; the descriptions below clarify the recognition problem represented by each benchmark.

Blood. BloodMNIST is an eight-class microscopy benchmark for recognizing peripheral blood-cell morphology. Its classes include granulocytes, lymphocytes, and other cells with visually similar staining patterns. Fine-grained morphology makes it a useful stress test for whether a merged classifier preserves class-specific directions rather than concentrating its predictions on a small subset of cells.

Derma. DermaMNIST contains seven dermoscopic skin-lesion categories. Its prevalence is strongly imbalanced and several categories share colour and texture patterns. This setting tests whether a merger distinguishes rare lesions without erasing the clinically meaningful prevalence of common lesions.

Organ-C and Organ-S. OrganCMNIST and OrganSMNIST provide 11-way abdominal CT organ recognition in coronal and sagittal views, respectively. The two views share the same broad task but change the visible anatomy and image geometry, testing robustness across CT planes.

Ultrasound. UltrasoundMNIST is an eight-class ultrasound-structure benchmark. Speckle noise, low contrast, and view-dependent boundaries make it particularly suitable for the prediction-collapse diagnostics and the per-class gap visualization in Fig. 5.

Detailed Ablation Configurations and Interpretation

All internal ablations use the same five datasets, four backbones, client counts, and Dirichlet settings as the main experiment. Thus, a change in Table 6 or Table 7 isolates one design choice while holding the remaining LAMP-Merge pipeline fixed.

Diagnostic Prototype Reconstruction For class c, DPR aggregates reference-space prototypes as

$$p_c = \sum_{i=1}^{K} \alpha_{i,c}\mu_{i,c}, \qquad \alpha_{i,c} = \frac{e_{i,c}}{\sum_j e_{j,c}}. \tag{10}$$

Table 9: Class-wise statistics and representative samples for Blood. The layout expands the corresponding row in the main-paper dataset table: the class-count field is replaced by individual class names. Samples are taken directly from the training split.

Dataset	Class	Train	Val	Test	Total	Sample
	Basophil	852	122	244	1,218	
	Eosinophil	2,181	312	624	3,117	
	Erythroblast	1,085	155	311	1,551	
	Immature granulocytes	2,026	290	579	2,895	
Blood	Lymphocyte	849	122	243	1,214	
	Monocyte	993	143	284	1,420	
	Neutrophil	2,330	333	666	3,329	
	Platelet	1,643	235	470	2,348	

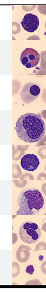


Table 10: Class-wise statistics and representative samples for Derma, using the same format as Table 9.

Dataset	Class	Train	Val	Test	Total	Sample
	Actinic keratosis / carcinoma	228	33	66	327	
	Basal cell carcinoma	359	52	103	514	
	Benign keratosis-like lesion	769	110	220	1,099	
Derma	Dermatofibroma	80	12	23	115	
	Melanoma	779	111	223	1,113	
	Melanocytic nevus	4,693	671	1,341	6,705	
	Vascular lesion	99	14	29	142	

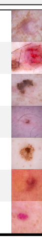


Table 11: Class-wise statistics and representative samples for Organ-C.

Dataset	Class	Train	Val	Test	Total	Sample
	Bladder	1,148	191	828	2,167	
	Femur–left	619	102	431	1,152	
	Femur–right	595	96	421	1,112	
	Heart	600	202	421	1,223	
	Kidney–left	1,088	132	727	1,947	
	Kidney–right	1,170	157	735	2,062	
Organ-C	Liver	2,986	429	1,835	5,250	
	Lung–left	1,002	347	549	1,898	
	Lung–right	1,022	352	557	1,931	
	Pancreas	1,173	179	750	2,102	
	Spleen	1,572	205	962	2,739	

$$e_{i,c} = (n_{i,c} + 1)^{\gamma} \mathbf{1}[n_{i,c} > 0]. \tag{11}$$

Here $\mu_{i,c}$ is client i's class-conditional reference-space mean and $n_{i,c}$ is its support for that class. The substitutions below test whether the class semantics in $\mu_{i,c}$ and the class-specific reliability in $\alpha_{i,c}$ are both required.

Table 12: Class-wise statistics and representative samples for Organ-S.

Dataset	Class	Train	Val	Test	Total	Sample
	Bladder	1,148	188	811	2,147	
	Femur–left	630	104	439	1,173	
	Femur–right	614	95	445	1,154	
	Heart	721	246	510	1,477	
	Kidney–left	1,132	140	704	1,976	
	Kidney–right	1,119	159	693	1,971	
Organ-S	Liver	3,464	491	2,078	6,033	
	Lung–left	741	261	397	1,399	
	Lung–right	803	275	439	1,517	
	Pancreas	2,004	280	1,343	3,627	
	Spleen	1,556	213	968	2,737	

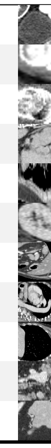


Table 13: Class-wise statistics and representative samples for Ultrasound. The source benchmark retains anonymized class identifiers, which are reported without assigning unverified clinical names.

Dataset	Class	Train	Val	Test	Total	Sample
	Class 0	420	60	121	601	
	Class 1	419	59	121	599	
	Class 2	408	58	117	583	
	Class 3	630	90	180	900	
Ultrasound	Class 4	345	49	100	494	
	Class 5	489	69	141	699	
	Class 6	671	95	193	959	
	Class 7	487	69	140	696	

Classifier-head aggregation (CHA). CHA replaces the class prototype in Eq. 10, $\mu_{i,c} \longrightarrow h_{i,c}$, with the local classifier-head direction. It tests whether client heads already provide a common, class-aligned representation. Its mean ACC is 25.63%, a 36.58-point reduction from LAMP-Merge (62.21%), showing that independently trained head directions are not a reliable substitute for shared-space class evidence.

Global-feature mean (GFM). GFM replaces the class-conditional prototype by the same class-agnostic feature mean for every class, $\mu_{i,c} \longrightarrow g$. This tests whether generic client appearance alone can drive merging. Its 26.50% mean ACC is 35.71 points below LAMP-Merge, indicating that class conditioning, not merely feature aggregation, is essential.

Support-only synthetic head (SSH). SSH replaces the aggregated prototype $p_c = \sum_i \alpha_{i,c}\mu_{i,c}$ by $p_c \longrightarrow u_c$, where u_c is a fixed random unit vector. It preserves the classifier interface while eliminating diagnostic content. The 10.89% mean ACC, 51.32 points below LAMP-Merge, verifies that performance does not arise from support counts or the cosine-head form alone.

Shuffled-label prototype (SLP). SLP replaces the class-aligned aggregate by $p_c \longrightarrow p_{\sigma(c)}$, for a permutation σ of the diagnosis labels. Its 19.57% mean ACC (a 42.64-point reduction) directly tests semantic alignment: useful class evidence must be assigned to the correct output class.

Uniform client weight (UCW). UCW replaces the support-aware weight $e_{i,c} = (n_{i,c}+1)^{\gamma}\mathbf{1}[n_{i,c}>0]$ by equal class-presence evidence:

$$e_{i,c} \longrightarrow \mathbf{1}[n_{i,c}>0], \qquad \alpha_{i,c} = \frac{e_{i,c}}{\sum_j e_{j,c}}. \tag{12}$$

It tests whether observing a class is sufficient, independent of the amount of evidence. Its 58.33% mean ACC is 3.88 points below LAMP-Merge, showing that support magnitude provides useful reliability information beyond class presence.

Binary support only (BSO). BSO replaces both class support and prevalence counts by presence indicators, $n_{i,c}, m_{i,c} \longrightarrow \mathbf{1}[n_{i,c}>0]$. It tests whether fine-grained class frequency is necessary in both modules. The mean ACC falls to 55.18% (7.03 points below LAMP-Merge), the largest loss among the weighting controls.

Global client-size weight (GCSW). GCSW replaces class-specific support in the reliability weight by total client size:

$$e_{i,c} = (n_{i,c}+1)^{\gamma}\mathbf{1}[n_{i,c}>0] \longrightarrow (N_i+1)^{\gamma}\mathbf{1}[n_{i,c}>0]. \tag{13}$$

Here $N_i = \sum_k n_{i,k}$ is the total support of client i. GCSW tests whether a generally large client can stand in for strong evidence of a particular class. Its 59.39% mean ACC, 2.82 points below LAMP-Merge, shows that local data volume should be assigned at the class level.

DPR summary. The semantic controls cause large reductions (35.71–51.32 points), while the weighting controls cause smaller but consistent reductions (2.82–7.03 points). Together, these results support the two parts of Eq. 10: class-conditioned shared-space evidence provides the discriminative direction, and class-specific support determines how reliably each client contributes to that direction.

Long-tail Prevalence Calibration LPC estimates an empirical prior and adds a centered, gated log-prior bias after DPR:

$$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}}, \qquad b_c = \lambda\left(\log \pi_c - \frac{1}{C}\sum_k \log \pi_k\right). \tag{14}$$

$$\ell_c(x) = w_c^{\top}\phi_0(T(x)) + \mathbf{1}[r>\tau]b_c. \tag{15}$$

The three controls retain DPR and test the prior estimator or the activation gate in Eq. 14.

Uniform prevalence prior (UPP). UPP replaces the empirical prior in Eq. 14 by $\pi_c \longrightarrow 1/C$. It tests whether calibration should ignore the observed class distribution. Mean ACC is 58.91%, 3.30 points below LAMP-Merge, demonstrating that the empirical prior is useful rather than a source of collapse.

Always-on calibration (AOC). AOC replaces the activation gate in the score $\ell_c(x)$ by $\mathbf{1}[r>\tau] \longrightarrow 1$. It tests whether the prior should be applied uniformly to every configuration. Its 61.90% mean ACC is 0.31 points below LAMP-Merge. The small aggregate gap indicates that calibration is broadly useful, while the gated design avoids imposing a prior when a configuration is not sufficiently long-tailed.

Client-balanced prior (CBP). CBP replaces the sample-count empirical prior by equal client influence:

$$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}} \longrightarrow \frac{1}{K}\sum_i \frac{m_{i,c}}{\sum_k m_{i,k}}. \tag{16}$$

It tests whether each client should contribute equally regardless of sample volume. The 58.97% mean ACC, 3.24 points below LAMP-Merge, favors the sample-count empirical prior over a client-balanced estimate.

LPC summary. Every LPC control is below the 62.21% LAMP-Merge mean. UPP and CBP show that both prevalence information and its sample-size weighting matter; AOC shows that the activation criterion is a targeted safeguard rather than the primary source of the gain. Hence LPC calibrates DPR's restored class directions instead of replacing them with a prior.

Additional Analysis of DPR and LPC

Prototype estimation and class margins Let $\mu_c^{\star}$ be the population class mean in the shared reference space. For client-level class estimates with bounded second moments, the mean-squared error of the aggregated prototype obeys

$$\mathrm{E}\|p_c - \mu_c^{\star}\|_2^2 \le \sum_i \alpha_{i,c}^2 \frac{\sigma_c^2}{n_{i,c}} + \mathrm{Bias}_c^2. \tag{17}$$

The first term is a variance contribution: it decreases with support $n_{i,c}$, but only clients that observed class c may contribute. The second term represents systematic mismatch between a local class estimate and the shared-space population mean. Equation 17 explains both support-related controls: UCW ignores unequal estimation variance, whereas GCSW confuses total client size with the support relevant to class c.

For a reference feature $z = \phi_0(T(x))$, consider two candidate classes c and d. The population margin is

$$\Delta_{c,d}(x) = (\mu_c^{\star})^{\top} z - (\mu_d^{\star})^{\top} z. \tag{18}$$

If

$$\|p_c - \mu_c^{\star}\|_2 + \|p_d - \mu_d^{\star}\|_2 < \frac{\Delta_{c,d}(x)}{\|z\|_2}, \tag{19}$$

then the reconstructed prototypes preserve the ordering between c and d. Thus, class-specific aggregation is not intended to reproduce arbitrary local heads: it reduces prototype error sufficiently to retain discriminative class margins. The large losses for CHA, GFM, SSH, and SLP are consistent with breaking precisely this semantic margin mechanism.

Bounded prevalence correction The centering in Eq. 14 removes the uniform component of the log prior. Therefore only relative scores change, with

$$b_c - b_d = \lambda \log \frac{\pi_c}{\pi_d}. \tag{20}$$

The gate $\mathbf{1}[r > \tau]$ applies this relative correction only under sufficient prevalence imbalance, and the finite λ bounds its magnitude. DPR consequently supplies the class directions, while LPC adjusts the decision boundary only when the empirical class prevalence provides relevant evidence. This division also explains why a prevalence-only mechanism cannot solve the semantic controls in Table 6.

Metric Definitions

Let y_n and $\hat{y}_n$ denote the true and predicted labels of sample n, respectively. Let $q_c = N^{-1} \sum_n \mathbf{1}[\hat{y}_n = c]$ be the predicted class distribution and $p_c = N^{-1} \sum_n \mathbf{1}[y_n = c]$ the true distribution. The prediction-collapse diagnostics are

$$\rho = \max_c q_c, \qquad C_{\text{eff}} = \exp\left(-\sum_c q_c \log q_c\right). \tag{21}$$

$$\mathrm{TV}(q, p) = \frac{1}{2} \sum_c |q_c - p_c|. \tag{22}$$

The collapse ratio ρ measures the largest predicted-class share, C_{eff} measures the effective number of classes used by the predictor, and TV measures distributional disagreement with the labels. Lower ρ and TV and higher C_{eff} are preferred. The absolute per-class gaps in Fig. 5 are $|q_c - p_c|$ in percentage points and expose which classes contribute to TV.

Hyperparameter-Sweep Reproducibility

Every point in Fig. 6 is a completed full-scope experiment covering five datasets, four backbones, three client counts, and three Dirichlet settings. The plotted curves retain measured points in ascending parameter order; no smoothing, interpolation, or best-run selection is applied. The LPC curves combine the completed $\tau \in \{2.0, 2.5, 3.0\}$ grid with newly completed $\tau \in \{1.5, 3.5\}$ sweeps; the calibration strengths span $\lambda \in \{3.00, 3.25, \ldots, 5.25\}$, with $\gamma = 0.55$ and $s = 18.75$ fixed. DPR completion records are plotted only after the full grid point has succeeded. The selected operating point is $\gamma = 0.55$, $s = 18.75$, $\tau = 2.5$, and $\lambda = 4.25$, making its reported value traceable to the same completed grid as its neighboring points.